

Shreya Rose Jimson

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EDUCATION

PES University <i>Bengaluru, Karnataka, India</i>	B.Tech - Electronics and Communication Engineering, CGPA: 8.54/10 <i>August 2023 – May 2027</i>
Relevant Courses: Semiconductor Devices and Fabrication (UE23EC343BB4), Analog Circuit Design (UE23EC252A), Electronic Principles And Devices (UE23EC141B), Low Power VLSI (UE23EC342BB4), Transmission Lines, Waveguides and Antennas (UE23EC351B), Electromagnetic Field Theory (UE23EE242B), Machine Learning and Applications (UE23EC352B), and Artificial Neural Networks (UE23EC343AC1)	
CMR NATIONAL PU College <i>Bengaluru, Karnataka, India</i>	Class 12, Karnataka Pre University Board, Score: 95.2% <i>June 2021 – May 2023</i>
Gopalan National School <i>Bengaluru, Karnataka, India</i>	Class 10, Indian Certificate of Secondary Education (ICSE), Score: 96.7% <i>June 2008 – May 2021</i>

PROJECTS

BSIM-CMG Compact Modeling and Circuit Verification of 3-Channel NSFET | *Ongoing*

- Integrated physical device dimensions from a Sentaurus TCAD 3-channel Nanosheet FET (NSFET) into a BSIM-CMG Verilog-A framework to enable advanced circuit-level simulation in Cadence Virtuoso
- Calibrated threshold voltage, subthreshold swing, and saturation current by designing a factorial experiment and applying linear least squares fitting in Python to automate physical parameter extraction from DC analysis traces
- Validated the calibrated compact model's robustness and logic-level functionality by designing, simulating, and verifying a resistive load inverter and an SRAM cell
- Currently extending the device model to investigate self-heating effects (SHE) and their impact on electrothermal performance, mobility degradation, and thermal reliability

TCAD Modeling and Characterization of Si/SiGe Heterojunction Line TFET

- Developed a Sentaurus TCAD model of a Si/SiGe heterojunction Line Tunnel Field-Effect Transistor (L-TFET) based on published IEEE research
- Simulated and verified transfer characteristics, analyzing the impact of electric field distribution and tunneling phenomena on device performance

TCAD Modeling and Characterization of Gate-All-Around (GAA) Nanowire FET

- Developed a 3D Sentaurus TCAD model of a sub-5 nm Gate-All-Around (GAA) Nanowire FET based on published IEEE research
- Implemented carrier transport, mobility, and electrostatic models to study nanoscale device behavior and short-channel effects
- Simulated transfer (ID-VGS) characteristics and extracted key performance metrics including threshold voltage, ON/OFF current ratio, and transconductance

Semiconductor Manufacturing Failure Prediction using Support Vector Machine (SVM)

- Built a Semiconductor Manufacturing Failure Prediction system using Machine Learning on the SECOM dataset to identify defective manufacturing processes
- Engineered an end-to-end data pipeline incorporating robust preprocessing, SMOTE class balancing, and PCA, subsequently optimizing an SVM model via GridSearchCV to achieve enhanced failure detection validated by Accuracy, Recall, and ROC-AUC metrics.

Analog Low Power CMOS Full Adder Design in 45nm Technology

- Designed a 45nm Hybrid 10T CMOS Full Adder in Cadence Virtuoso, utilizing supply voltage scaling, HVT transistors, and glitch reduction to minimize dynamic and leakage power
- Validated transistor-level functionality and confirmed significant power reduction via transient simulation, waveform verification, and power analysis across all input combinations

EXPERIENCE

Research Intern

May 2026 – Present

Chips to Start-up (C2S), PES University

Bengaluru, Karnataka, India

- Characterized a sub-5nm GAA Nanowire FET in Sentaurus TCAD and translated the extracted physical data into a BSIM-CMG compact model, automating parameter calibration via Python least-squares fitting and validating SRAM logic
- Currently investigating self-heating effects (SHE) within the device to analyze the impact on electrothermal performance, mobility degradation, and overall thermal reliability

Research Intern

June 2025 – August 2025

Crucible Of Research and Innovation (CORI), PES University

Bengaluru, Karnataka, India

- Developed a DETR-based pedestrian detection system, implementing the end-to-end deep learning pipeline in PyTorch
- Validated the hybrid CNN-Transformer architecture via forward-pass tracing, tensor analysis, and code-level debugging

Research Intern

June 2024 – August 2024

Center for Robotics, Automation and Intelligent Systems (cRAIS) PES University

Bengaluru, Karnataka, India

- Simulated quadcopter dynamics and autonomous path navigation in MATLAB/Simulink with Simscape, conducting parameter sweeps to evaluate the effects of package mass and target speed on performance

SKILLS

Engineering Software & CAD: Synopsys Sentaurus TCAD, Cadence Virtuoso, LTspice, Simulink

Programming and Tools: Python, LaTeX, MATLAB

Core Competencies: Compact Modeling (BSIM-CMG), Analog Circuit Analysis and Testing, Device Characterization, Semiconductor Device Physics

Soft Skills: Leadership, Cross-Functional Collaboration, Adaptability, Problem-Solving, Creativity, Communication, Teamwork, Organized, Strong Interpersonal skills, Proactive, Fast Learner, Hardworking, Curious, Precision and Attention to Detail, Highly Committed, Resilient

PUBLICATIONS

Design and Evaluation of A Real-Time Pedestrian Detection System Using Vision Transformer

IEEE 17th International Conference on Computational Intelligence and Communication Networks (CICN)

HONOURS AND AWARDS

- Prof. CNR Rao Scholarship (Semester V - Merit Scholarship for top performers of the semester)
- Distinction Award Certificate (DAC) Scholarship (Semester I, II, III, IV)

CERTIFICATIONS

- Summer School 2026 - Centre for Nano Science and Engineering (CeNSE), IISc (June 2026)
- India Semiconductor Workforce Development Program (ISWDP) Samsung Student Fellowship Grade II recipient - Cohort 6 (August - December 2025)
- Electrical Engineering Summer School - Indian Institute of Science (July 2025)
- Semiconductor Fabrication 101 - Purdue University (February 2026)
- Quantum Technology Workshop - Quantum Research Park (QuRP), IISc (June 2024)

REFERENCES

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